Roll Number: _____

Thapar Institute of Engineering & Technology, Patiala

Department of Computer Science and Engineering

Mid Semester Test (MST)

B.E. (4th Semester)	Course Code: UTA018 Course Name: Object Oriented Programming
Date: March 10, 2026	Time: 11:30-13:30
Duration: 2 Hours, M. Marks: 30	Name of Faculty: Dr. Saurabh, Dr. Harpreet

Note: Attempt all questions in order. Assume missing data, if any, suitably.

		Marks	CO/ BL
Q1	(a) Analyze the various types of inheritance available in object-oriented programming, explain their syntactic structure, and illustrate each with a clear diagram. Additionally, describe four advantages of inheritance. (b) A software company is developing a system for a university. There is a base class Person that contains common attributes such as name and id. Two classes, Student and Employee, inherit from Person. Now, a new class TeachingAssistant is created that inherits from both Student and Employee, since a teaching assistant is both a student and an employee. During implementation, the developers observe that accessing the name attribute from a TeachingAssistant object results in ambiguity. Explain why this ambiguity occurs in the above inheritance structure. How would you modify the inheritance design to eliminate this problem?	(3+3)	2/2
Q2	(a) Are all the functions defined inside a class inline? Define 4 cases when functions cannot be treated as inline functions. (b) Design two classes named Circle and CircleManager. Declare CircleManager as a friend of the Circle class. Implement a member function in CircleManager that doubles the radius of a Circle object. Demonstrate how declaring friendship enables CircleManager to directly access and modify the private data members of the Circle class.	(3+3)	1/2
Q3	(a) A vehicle rental company manages different types of vehicles such as cars and bikes. While all vehicles share some common properties, each type of vehicle has its own specific features. Design and implement a C++ program for a Vehicle Rental System which has the following requirements: Create a base class Vehicle with: Protected data members: vehicleID, brand, and a private data member: rentalRate. Provide public member functions to: Accept vehicle details, Display vehicle information. Create a class Car that publicly inherits from Vehicle. Add an additional data member numberOfSeats. Provide a function to display car details. Create another class Bike that also inherits from Vehicle. Add a data member engineCapacity. Create a class ElectricCar that inherits publically from the Car class. Add a data member batteryCapacity. Write a main function to create objects of Car, Bike, ElectricCar, and input and display their details and the functions to calculate the cost of hiring a car or a bike. Initialize the objects using parameterized constructors. If needed, additional data members can be added. (b) Can friend functions and static members functions be inherited? Give reasons	(4+2)	2/3
Q4	(a) A book shop maintains the inventory of books that are being sold at the shop. The list includes details such as author, title, price, publisher and stock position. Whenever a customer wants a book, the salesperson, inputs the title and author and the system searches the list and displays whether it is available or not. If it is not, appropriate message is displayed. If it is, then the system displays the book details and requests for the number of copies required. If the requested copies are available, the total cost of the requested copies is displayed; otherwise the message "required copies not in stock is displayed". Design a system using a class called books with suitable member functions, constructors and destructors. Use dynamic constructor to enter the book information and to allocate the	(5+1)	1/2

	memory space required. The search of each book whether it is successful or not is recorded as a transaction and the total number of transactions in a system should be recorded. (b) Define following: Static objects, and Abstraction.		
Q5	(a) Below is the partially completed program to count the number of objects which are created in the main function. There are multiple blanks in the program which you need to fill with correct option and also specify the output of this program.	(3)	1/3

```
#include <iostream>
using namespace std;
class A {
private:
   _Blank 1_ int count;
public:
   A ( ) { _Blank 2_++; }
   static int getCount ( ) { return _Blank 3_; }
};
int A::count= 5;
int main( ) {
   A c1, c2, c3, c4;
   cout <<"Total objects created: " << _Blank 4_ << endl;
   return 0;
}
```

(b) Fill the three blanks in the following program, if the output is 11.

(3)

```
#include <iostream>
using namespace std;
class Test {
private:
   int value;
public:
   Test(int val) : value(val){ }
   void setValue (int val) {
      this->value = ++val;
   }

_Blank 1_ increase( ) {
      this->value += 2;
      return _Blank 2_;
   }
   void display( ) {
      cout << "Value: " << value << endl;
   }
};
int main( ) {
   Test P(4);
   P.setValue(_Blank 3_);
   P. increase( ). increase( ). display( );
   return 0;
}
```